ENERGY ON ONE PAGE

Energy: from stores to stationary action

The sum $T+V$ is conserved, the difference $T-V$ is optimised, and the virial theorem ties the two together.

1 / THE RULE AND THE PICTURE

Nature's ledger is read at three levels, and the last gives two books

Stores
a number that balances
E_{total} constant
→ middle school

Sum
$H = T + V$
conserved
→ GCSE and A-level

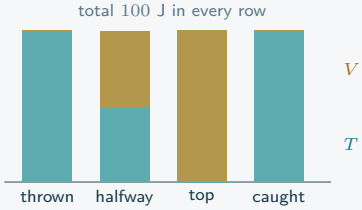
Difference
$L = T - V$
$S = \int L\,dt$ stationary
→ university

The sum says what is preserved; the difference says which path is taken:

$$H = T + V, \qquad L = T - V$$

2 / STORES AND THE LEDGER / MIDDLE SCHOOL

Energy is a number that always balances, and its quality can still fall



A 0.50 kg ball thrown up at $20\ \mathrm{m\,s^{-1}}$ starts with $T = \frac{1}{2}mv^2 = 100$ J, passes the half-way row with 50 J in each store and reaches 20.4 m with all 100 J in $V = mgh$; every row sums to the same total. The stores are bookkeeping, and the exchange rates are $E = mc^2$, $E = hf$ and $E = k_B T$. Quality still falls: a 500 nm photon of 2.48 eV from the Sun returns as thirty $15\,\mu$m photons of 0.083 eV, the same total spread over thirty lines.

3 / THE SUM / GCSE AND A-LEVEL

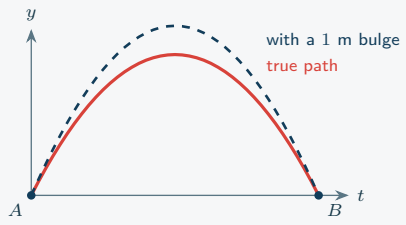
Work done changes T, and the sum $E = T + V$ stays constant

step	statement	reason
1	$W = Fs$	definition of work
2	$W = mas$	Newton's second law, $F = ma$
3	$as = \frac{1}{2}(v^2 - u^2)$	the suvat relation $v^2 = u^2 + 2as$
4	$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta T$	the work–energy theorem

Potential energy is the energy of position, $V = mgh$ near the surface and $V(r) = -GMm/r$ in general, the minus sign meaning that bringing masses together releases energy; the conserved sum $E = T + V$ is the Hamiltonian.

4 / THE DIFFERENCE / STATIONARY ACTION

Of all paths from A to B nature takes the one that makes $S = \int (T - V)\,dt$ stationary

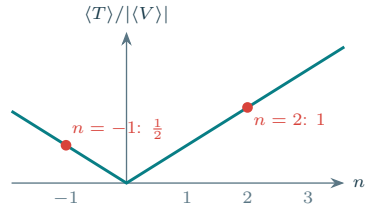


For a 1 kg ball going from the ground at $t = 0$ back to the ground at $t = 2$ s, the true parabola gives $S = -32.01$ J s, while adding a 1 m sinusoidal bulge raises it to -30.78 J s, and in general $S(\epsilon) = S_0 + \frac{\pi^2}{8}\epsilon^2$, so the first variation is zero and the true path is the stationary one. Varying the path with fixed ends and integrating by parts turns $\delta S = 0$ into the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = 0 \quad \Longleftrightarrow \quad m\ddot{r} = -\frac{\partial V}{\partial r}$$

5 / THE VIRIAL THEOREM / THE SUM AND THE DIFFERENCE MEET

For $V \propto r^n$ the time averages obey $2\langle T\rangle = n\langle V\rangle$



The derivation differentiates $G = \sum \mathbf{p}\cdot\mathbf{r}$, and since $\langle \dot{G}\rangle = 0$ on a bound orbit, $2\langle T\rangle = -\langle \sum \mathbf{F}\cdot\mathbf{r}\rangle = n\langle V\rangle$. For gravity or Coulomb, $n = -1$, so $\langle T\rangle = -\frac{1}{2}\langle V\rangle$ and $E = \frac{1}{2}\langle V\rangle = -\langle T\rangle < 0$: in hydrogen $\langle T\rangle = 13.6$ eV, $\langle V\rangle = -27.2$ eV and $E = -13.6$ eV, so of the 27.2 eV of potential energy released, half becomes the electron's motion and half leaves as the photon, and gravitational collapse likewise heats half and radiates half. The oscillator, $n = 2$, gives equipartition, and $n = -1$ and $n = 2$ are the two Bertrand potentials with closed orbits.

$$E = \left(\tfrac{n}{2} + 1\right)\langle V\rangle, \qquad \text{binding energy} = -E = \langle T\rangle \quad (n = -1)$$

ASIDE / THE THREAD

The sum is what nature keeps and the difference is what nature optimises; for bound motion in a power law both follow from the one number n.

ASIDE / CHECK IN PYTHON

quad of $\frac{1}{2}\dot{y}^2 - gy$ over $[0, 2]$ gives -32.013 for the parabola and -30.780 with the bulge.

Convention. After the Energy from stores to stationary action page; $g = 9.8\ \mathrm{m\,s^{-2}}$; $\langle\cdot\rangle$ is a time average over a bound orbit. Follows the binding-energy sheet.

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